Acute Medical Diseases (Critical Care)

Pain, agitation, delirium • DKA/HHS • sepsis • shock • cardiac arrest • acid–base — all key points for the exam (SCCM PADIS, SSC 2021, AHA ACLS, ADA 2024) — drug table at the end

1. Pain, Agitation/Sedation, Delirium, Immobility and Sleep (PADIS)

ICU patients commonly have **pain** (procedures, tubes, surgery), **agitation**, and **delirium** — they are linked: untreated pain → agitation → delirium. Core principles: **assess with validated tools**, **treat pain first (analgesia-first / analgosedation)**, use **light sedation**, **avoid benzodiazepines**, and use the **ABCDEF bundle**.

ABCDEF bundle

Letter	Element
A	**Assess, prevent, and manage pain** (NRS, BPS, CPOT)
B	**Both** spontaneous awakening trials (**SAT**) and spontaneous breathing trials (**SBT**) — daily, coordinated
C	**Choice** of analgesia and sedation (light sedation, non-benzodiazepine)
D	**Delirium**: assess, prevent, and manage (CAM-ICU, ICDSC)
E	**Early mobility** and exercise
F	**Family** engagement and empowerment

A. Pain assessment

- **Self-report = gold standard**: **NRS 0–10** (visual format in ICU) • goal usually **< 4**.
- Unable to self-report (intubated, sedated) with intact motor function: **Behavioral Pain Scale (BPS**, 3–12; **> 5** = significant pain) or **Critical-Care Pain Observation Tool (CPOT**, 0–8; **> 2** = pain).
- **Vital signs alone are not valid** for pain assessment (only a cue to assess further).
- **Preemptive analgesia** before painful procedures (turning, chest tube removal, suctioning, wound care).

Pain treatment

- **Opioids = first-line** for non-neuropathic pain • IV push or infusion • **no opioid is superior** → choose by organ function, hemodynamics, and duration.
- **Multimodal (opioid-sparing) adjuncts: acetaminophen** (IV/PO), **low-dose ketamine** (post-surgical), **neuropathic agents** (gabapentin, pregabalin, carbamazepine — e.g., Guillain-Barré, neuropathy), **nefopam** (outside US) • NSAIDs and IV lidocaine **not routinely** recommended (AKI, bleeding).
- **Analgosedation** = analgesic first, then add sedative only if needed → ↓ sedative exposure.

Opioid	IV dose	Key points
Fentanyl	Push 0.35–0.5 mcg/kg q30–60 min (25–100 mcg) • infusion 0.7–10 mcg/kg/h (25–200 mcg/h)	**Fastest onset (1–2 min)** • **no histamine release → hemodynamically stable** • preferred in **renal failure** • lipophilic → **accumulates** with prolonged infusion • **CYP3A4** substrate • **chest wall rigidity** (high/rapid doses) • serotonin syndrome risk
Hydromorphone	Push 0.2–0.6 mg q1–2h • infusion 0.5–3 mg/h	~5–7× more potent than morphine • usable in renal impairment (safer than morphine; H3G may accumulate) • **dosing errors** (confused with morphine)
Morphine	Push 2–4 mg q1–2h • infusion 2–30 mg/h	**Histamine release → hypotension, pruritus, bronchospasm** • active metabolites (**M6G**, M3G) **accumulate in renal failure → avoid in AKI/CKD**

Opioid	IV dose	Key points
Remifentanil	Load 1.5 mcg/kg • infusion 0.5–15 mcg/kg/h	**Plasma esterase** metabolism → ultra-short (3–10 min) • **no accumulation** (organ independent) • good for frequent neuro exams • hyperalgesia/rapid withdrawal after stopping • dose on IBW
Methadone	PO/IV 10–40 mg q6–12h	Used to **wean** long opioid infusions • NMDA antagonism • **QT prolongation** • long variable half-life
Meperidine	—	**Avoid**: normeperidine → **seizures** (renal failure), serotonin syndrome with MAOIs, delirium

- **Opioid ADRs: respiratory depression**, hypotension, **constipation**/ileus (give **prophylactic bowel regimen** — senna/PEG; methylnaltrexone for refractory opioid-induced constipation), nausea, urinary retention, pruritus, delirium, tolerance, **withdrawal** after > 7 days (taper 10–20% per day; methadone or clonidine help).
- **Reversal: naloxone 0.04–0.4 mg IV** titrated every 2–3 min (lower doses avoid abrupt withdrawal, pain, pulmonary edema) • short half-life → may need repeat/infusion.
- **Acetaminophen IV:** 650 mg q4h or **1 g q6h (max 4 g/day; ≤ 2 g/day in liver disease/malnutrition/alcohol use)** • can cause hypotension (IV) • hepatotoxicity.
- **Ketamine (analgesic):** 0.5 mg/kg IV load then **1–2 mcg/kg/min** (0.06–0.12 mg/kg/h) → opioid-sparing after surgery.

B. Agitation and sedation

- **First look for causes of agitation: pain**, **hypoxia**, hypercapnia, **hypoglycemia**, **delirium**, **withdrawal** (alcohol, nicotine, opioids, benzodiazepines), full bladder, ventilator dyssynchrony, anxiety, sleep deprivation.
- **Assessment: RASS** (Richmond Agitation-Sedation Scale, **−5 to +4**; goal **light sedation −2 to 0**) or **SAS** (Riker, 1–7; goal 3–4).
- **Light sedation** (patient calm, arousable, follows commands) → ↓ ventilator days, ICU stay, delirium, PTSD • deep sedation only for specific indications (severe ARDS on neuromuscular blockers, refractory status epilepticus, ↑ ICP, refractory ventilator dyssynchrony).
- **Daily sedation interruption (SAT)** or **nurse-protocolized light sedation** paired with SBT.
- **Propofol or dexmedetomidine are preferred over benzodiazepines** (midazolam, lorazepam) in mechanically ventilated adults (shorter time to extubation, ↓ delirium).
- **Benzodiazepines still preferred for: alcohol/benzodiazepine withdrawal**, **seizures/status epilepticus**, need for deep sedation/amnesia (e.g., with paralytics), palliative sedation, contraindications to propofol/dexmedetomidine.

Sedative	Dose	Key points
Propofol	Start **5 mcg/kg/min** → ↑ 5–10 every 5–10 min • usual **5–50 mcg/kg/min** (max ~80) • loading not recommended (hypotension)	GABA-A agonist • onset 1–2 min, **rapid offset** (neuro checks) • **no analgesia** • ADRs: **hypotension**, respiratory depression, bradycardia, **hypertriglyceridemia** (check TG at baseline and after 48 h; hold if > 400–500), **pancreatitis**, **PRIS**, green urine/hair, injection pain • **lipid emulsion: 1.1 kcal/mL** (count in nutrition) • strict aseptic technique (change tubing/vial every 6–12 h) • caution egg/soy allergy
Dexmedetomidine	**0.2–1.5 mcg/kg/h** (load 1 mcg/kg over 10 min usually omitted)	**Central α2 agonist** • **arousable** "cooperative" sedation • **no respiratory depression** → can use in non-intubated patients and during extubation • mild analgesic, opioid-sparing • ↓ delirium • ADRs: **bradycardia**, **hypotension** (load → transient hypertension), withdrawal (agitation, tachycardia, HTN) after prolonged use • **not for deep sedation or seizures** • hepatic metabolism
Midazolam	Push 0.01–0.05 mg/kg • infusion 0.02–0.1 mg/kg/h	Rapid onset (2–5 min) • **CYP3A4** (azoles, macrolides, diltiazem ↑) • active metabolite (**1-hydroxymidazolam**) **accumulates in renal failure** • lipophilic → accumulates in obesity/prolonged use • ↑ delirium

Sedative	Dose	Key points
Lorazepam	Push 0.02–0.04 mg/kg (≤ 2 mg) q2–6h • infusion 0.01–0.1 mg/kg/h (≤ 10 mg/h)	Glucuronidation (no active metabolites; OK in liver disease) • **propylene glycol toxicity** with high doses/infusions (> 1 mg/kg/day): **↑ osmolal gap (> 10–12)**, anion-gap **metabolic acidosis**, AKI, hemolysis • slower onset (5–20 min) • ↑ delirium
Ketamine	Load 0.1–0.5 mg/kg • infusion 0.05–0.4 mg/kg/h (sedation up to 1–2 mg/kg/h)	NMDA antagonist • **preserves BP and respiratory drive**, **bronchodilator** (status asthmaticus) • ADRs: **emergence reactions/hallucinations**, ↑ HR/BP, ↑ secretions, ↑ ICP (minor)

Propofol-related infusion syndrome (PRIS): high doses (**> 70–80 mcg/kg/min**) for **> 48 h**, especially with catecholamines/steroids, young patients → **metabolic (lactic) acidosis**, **rhabdomyolysis** (↑ CK), **hyperkalemia**, **hypertriglyceridemia**, **AKI**, **arrhythmias/bradycardia**, cardiac failure → **stop propofol immediately**, supportive care (pacing, dialysis/ECMO). Monitor CK, lactate, TG, K.

Neuromuscular blockers (NMBs)

- **No sedative or analgesic properties → always ensure deep sedation and analgesia first** (awareness during paralysis is traumatic) • monitor with **train-of-four** (goal 1–2 twitches of 4) + clinical assessment.
- Indications: facilitate intubation, **early severe ARDS** (P/F < 150 — intermittent boluses preferred over routine infusion), ventilator dyssynchrony, ↑ ICP, therapeutic hypothermia shivering.
- **Cisatracurium** (0.1–0.2 mg/kg then 1–3 mcg/kg/min): **Hofmann elimination** (organ independent) → preferred in **renal/hepatic failure** • **rocuronium** (fast, intubation 0.6–1.2 mg/kg) • **vecuronium** (active metabolite accumulates in renal failure).
- **Succinylcholine** (depolarizing, 1–1.5 mg/kg; intubation only): **hyperkalemia** → CI in **burns, crush injury, prolonged immobility, denervation/neuromuscular disease**, known hyperkalemia; **malignant hyperthermia** (treat with **dantrolene**); bradycardia.
- Complications: **ICU-acquired weakness** (↑ with corticosteroids, aminoglycosides), corneal abrasions (**eye lubricant**), pressure ulcers, **VTE** (prophylaxis), awareness.
- **Reversal: sugammadex** (rocuronium, vecuronium; ↓ hormonal contraceptive efficacy for 7 days) • neostigmine + glycopyrrolate.

C. Delirium

- **Definition:** acute disturbance of **attention and awareness** with **fluctuating** course + cognitive changes, caused by a medical condition/drug • ↑ **mortality**, ventilator days, LOS, **long-term cognitive impairment**.
- **Subtypes: hypoactive** (most common, often missed, worse prognosis), hyperactive (agitated), mixed.
- **Screen every shift: CAM-ICU** = **(1) acute change or fluctuating course** + **(2) inattention** + either **(3) altered level of consciousness (RASS ≠ 0)** or **(4) disorganized thinking** • or **ICDSC** (0–8; **≥ 4** = delirium).
- **Risk factors: modifiable** — **benzodiazepine use**, **blood transfusion**, deep sedation, immobility, sleep deprivation, restraints • nonmodifiable — **older age**, **dementia**, prior coma, emergency surgery/trauma, high illness severity (APACHE).
- **Deliriogenic drugs: benzodiazepines**, **anticholinergics** (diphenhydramine, hydroxyzine, promethazine, scopolamine, oxybutynin, TCAs), **opioids** (meperidine), **corticosteroids** (high dose), H2RAs (elderly), antiepileptics, dopamine agonists, fluoroquinolones, cefepime (renal failure).
- **Look for causes:** infection/sepsis, hypoxia, metabolic (Na, glucose, Ca), uremia, hepatic encephalopathy, **withdrawal**, urinary retention, constipation, pain.

Delirium prevention and treatment

- **Nonpharmacologic multicomponent prevention = mainstay:** reorientation, clocks/calendars, **eyeglasses and hearing aids**, **early mobilization**, **sleep hygiene** (lights off at night, noise reduction, cluster care), family presence, minimize restraints and tethers, avoid deliriogenic drugs.
- **No drug is recommended to PREVENT delirium** (haloperidol, atypical antipsychotics, dexmedetomidine, statins, ketamine — not for prevention).

- **Do not routinely use antipsychotics to treat delirium** (MIND-USA: haloperidol or ziprasidone did not shorten delirium or ↓ mortality).
- **Short-term antipsychotic** (haloperidol or atypical, e.g., quetiapine) is reasonable for **distressing symptoms**: anxiety, fear, hallucinations, delusions, or **agitation that endangers the patient/others** → **stop when symptoms resolve** (don't continue at transfer/discharge).
- **Dexmedetomidine** for **agitated delirium preventing weaning/extubation**.
- **Benzodiazepines worsen delirium** → avoid, **except alcohol/benzodiazepine withdrawal** (benzodiazepines ± phenobarbital).
- **Haloperidol:** 0.5–1 mg (elderly) to 2–5 mg IV/IM q20–30 min PRN • ADRs: **QTc prolongation/torsades** (baseline ECG; hold if QTc > 500 ms or ↑ > 60 ms), **EPS**, **NMS**, hypotension • correct K and Mg.
- **Quetiapine** 25–50 mg PO BID (↑ 50 mg/day; max 200 mg BID): sedation, orthostasis, QT, anticholinergic effects.

Immobility and sleep

- **Early mobilization/rehabilitation** → ↓ ICU-acquired weakness, ↓ delirium, ↓ ventilator days • safe when hemodynamically and respiratory stable.
- **Sleep:** nonpharmacologic (earplugs, eye masks, noise/light reduction, music) • **no recommendation for melatonin**; avoid propofol for sleep; dexmedetomidine at night (conditional) • assist-control ventilation at night.

Alcohol withdrawal syndrome (ICU)

- Onset 6–24 h after last drink • seizures 12–48 h • **delirium tremens 48–96 h** (autonomic instability, hallucinations, mortality) • CIWA-Ar (non-intubated).
- **Benzodiazepines = first-line** (symptom-triggered or front-loading): lorazepam (liver disease, elderly), diazepam, chlordiazepoxide • **phenobarbital** (adjunct or monotherapy for benzodiazepine-resistant) • dexmedetomidine as adjunct only (does not prevent seizures) • propofol for refractory/intubated.
- **Thiamine before glucose** (100–500 mg IV — Wernicke's) • folate, multivitamin, Mg, K, phosphorus repletion.

2. Diabetic Ketoacidosis (DKA) and Hyperosmolar Hyperglycemic State (HHS)

Feature	DKA	HHS
Typical patient	**T1DM** (also T2DM; SGLT2i users)	**Older T2DM**
Onset	Hours (< 24 h)	Days to weeks
Glucose	**≥ 200 mg/dL** (or prior diabetes) — can be < 250 (**euglycemic DKA**)	**≥ 600 mg/dL**
Ketones	**β-hydroxybutyrate ≥ 3.0 mmol/L** or urine ketones ≥ 2+	Minimal (< 3.0)
Acidosis	**pH < 7.3** and/or **HCO3 < 18**	pH ≥ 7.3, HCO3 ≥ 15
Anion gap	**↑ (> 10–12)**	Normal/mildly ↑
Osmolality	Variable (< 320)	**Effective osm > 300 mOsm/kg** (usually > 320)
Clinical	**Kussmaul breathing**, **fruity (acetone) breath**, N/V, **abdominal pain**, polyuria	**Severe dehydration** (~9 L), **altered mental status/coma**, focal deficits, seizures
Mortality	< 1–2%	5–20% (higher)

- **Pathophysiology:** insulin deficiency + ↑ counter-regulatory hormones (glucagon, cortisol, catecholamines) → hyperglycemia, **lipolysis → ketones** (DKA) → osmotic diuresis → dehydration and electrolyte losses (K, Na, phos, Mg) • in HHS, residual insulin prevents ketosis.
- **Precipitants: infection** (most common), **insulin omission**/pump failure, **new-onset diabetes**, MI, stroke, pancreatitis, trauma, alcohol/cocaine, pregnancy, **drugs**: **SGLT2 inhibitors**, **glucocorticoids**, **atypical antipsychotics**, thiazides, sympathomimetics, **immune checkpoint inhibitors**.

- **Labs:** glucose, BMP, **anion gap = Na − (Cl + HCO3)**, β-hydroxybutyrate, VBG, serum osmolality, CBC, phosphate, Mg, cultures, ECG (K), lipase, A1C.
- **Corrected Na** = measured Na + **1.6 × [(glucose − 100)/100]** • **effective osmolality** = 2 × Na + glucose/18.
- **Potassium paradox:** serum K often **normal or high** at presentation (acidosis + insulin deficiency shift K out of cells) despite **total body K deficit** → insulin drives K into cells → **hypokalemia** (main cause of arrhythmia/death).
- **Leukocytosis** common (stress) • amylase/lipase may be ↑ without pancreatitis • **pseudohyponatremia** from hyperglycemia.

Management (ADA/EASD 2024 consensus)

Step	Details
1. IV fluids (first)	**Isotonic crystalloid (0.9% NaCl or balanced: LR, Plasma-Lyte) 1–1.5 L in the 1st hour** (15–20 mL/kg) → then **250–500 mL/h** guided by volume status • switch to **0.45% NaCl** if corrected Na normal/high • **add dextrose 5–10%** when glucose **< 250 mg/dL (DKA)** or **< 300 mg/dL (HHS)** • HHS: slower correction (↓ osmolality ≤ 3–8 mOsm/kg/h)
2. Potassium (check before insulin)	**K < 3.5 → HOLD insulin**, give KCl **10–20 mEq/h** until K ≥ 3.5 • **K 3.5–5.0 → add KCl 10–20 mEq/h** to fluids (goal 4–5) • **K > 5.0 → no K**, recheck every 2 h
3. Insulin	**Regular insulin IV infusion 0.1 units/kg/h** (± bolus 0.1 units/kg) or **0.14 units/kg/h** without bolus • target glucose ↓ **50–75 mg/dL/h** • when glucose < 250 → **↓ to 0.05 units/kg/h** + dextrose, keep glucose **150–250** until DKA resolves • **mild–moderate DKA**: SC rapid-acting insulin (0.2 units/kg then 0.1 every 1 h, or 0.3 then 0.2 every 2 h) on a ward is acceptable • continue home basal insulin (optional)
4. Bicarbonate	**Only if pH < 7.0** → 100 mEq NaHCO3 in 400 mL water + 20 mEq KCl over 2 h; repeat until pH ≥ 7.0 (risk: hypokalemia, cerebral edema)
5. Phosphate	Not routine • give if **< 1.0 mg/dL** or with cardiac dysfunction, respiratory depression, hemolysis • K-phos 20–30 mEq (watch hypocalcemia)
6. Magnesium	Replace if low
7. Precipitant	Treat infection, MI, etc. • stop SGLT2i
Monitoring	Glucose **every 1 h** • electrolytes, VBG, β-hydroxybutyrate, anion gap **every 2–4 h** • fluid balance, mental status

- **DKA resolution: β-hydroxybutyrate < 0.6 mmol/L** + **venous pH ≥ 7.3** (or **HCO3 ≥ 18**) — anion gap closure also used (≤ 12) • **glucose normalizes before ketoacidosis resolves** → don't stop insulin when glucose is normal; add dextrose instead.
- **HHS resolution:** effective osmolality < 300, glucose < 250, urine output ≥ 0.5 mL/kg/h, cognitive recovery.
- **Transition to SC insulin:** when resolved and patient can eat → give **basal insulin SC 1–2 h before stopping the IV infusion** (overlap prevents rebound) • insulin-naive: **0.5–0.6 units/kg/day** (50% basal, 50% prandial) • known patients: resume home doses.
- **Complications of therapy: hypoglycemia**, **hypokalemia** (most dangerous), **cerebral edema** (mainly **children**; headache, ↓ consciousness → **mannitol 0.5–1 g/kg** or 3% saline), **hyperchloremic non-anion-gap acidosis** (large NS volumes — use balanced crystalloids), fluid overload/pulmonary edema, VTE (HHS — prophylaxis), rhabdomyolysis (HHS).
- **Euglycemic DKA** (SGLT2i, pregnancy, starvation, alcohol, low-carb diet): glucose < 250 but ketoacidosis → **dextrose early** with insulin; stop SGLT2i.
- **Alcoholic ketoacidosis:** fluids with **dextrose** + **thiamine first** — insulin usually not needed • **starvation ketosis**: mild acidosis, HCO3 rarely < 18.

3. Sepsis and Septic Shock

Definitions (Sepsis-3)

- **Sepsis** = **life-threatening organ dysfunction** caused by a **dysregulated host response to infection** • organ dysfunction = **↑ SOFA score ≥ 2** points.
- **Septic shock** = sepsis with **vasopressors required to keep MAP ≥ 65 mmHg AND lactate > 2 mmol/L** despite adequate fluid resuscitation (mortality > 40%).
- **qSOFA** (bedside, ≥ 2 = high risk): **RR ≥ 22**, **altered mentation** (GCS < 15), **SBP ≤ 100 • SSC 2021: do not use qSOFA alone** as a screening tool (vs SIRS, NEWS, MEWS).
- **SIRS** (older, ≥ 2): T > 38 °C or < 36 °C • HR > 90 • RR > 20 or PaCO2 < 32 • WBC > 12,000, < 4,000, or > 10% bands — sensitive, not specific.
- **SOFA** components: respiration (PaO2/FiO2), coagulation (platelets), liver (bilirubin), cardiovascular (MAP/vasopressors), CNS (GCS), renal (SCr/urine output).
- **Common sources: lung (pneumonia)** > abdomen > urinary tract > skin/soft tissue > catheter-related • older age, immunosuppression, diabetes, cancer ↑ risk.

Hour-1 bundle (Surviving Sepsis Campaign)

1. **Measure lactate → remeasure within 2–4 h** if initial > 2 mmol/L (lactate-guided resuscitation).
2. **Obtain blood cultures (≥ 2 sets) before antibiotics** — but **do not delay antibiotics** for cultures.
3. **Give broad-spectrum antibiotics.**
4. **Begin 30 mL/kg IV crystalloid** within the first 3 h for **hypotension or lactate ≥ 4 mmol/L**.
5. **Start vasopressors** if hypotensive during or after fluids to keep **MAP ≥ 65 mmHg**.

Antimicrobial therapy

- **Timing: possible septic shock or high likelihood of sepsis → within 1 hour** • possible sepsis without shock → rapid assessment, give within **3 hours** if infection concern persists • every hour of delay ↑ mortality in shock.
- **Empiric broad-spectrum** based on source, local antibiogram, prior cultures, recent antibiotics, healthcare exposure: **anti-pseudomonal β-lactam** (**piperacillin-tazobactam**, **cefepime**, **meropenem**)
- **Add MRSA coverage** (**vancomycin** or linezolid) if **high risk** for MRSA (prior MRSA, recent IV antibiotics, hemodialysis, IV drug use, skin/catheter source).
- **Two gram-negative agents** (β-lactam + aminoglycoside or fluoroquinolone) only if **high risk of MDR organisms** (prior MDR colonization, high local resistance) and septic shock — de-escalate once susceptibilities known.
- **Antifungal** (echinocandin — micafungin, caspofungin) if high risk of **Candida** (TPN, femoral line, broad antibiotics, abdominal surgery, Candida colonization, neutropenia).
- **Optimize PK/PD: full loading doses** regardless of renal function • **prolonged/extended infusion β-lactams** for maintenance (↑ time above MIC) • vancomycin AUC-guided dosing (AUC 400–600) • aminoglycosides once daily (high peak).
- **Source control** (drain abscess, remove infected catheters, debride) as soon as practical (within 6–12 h).
- **Daily de-escalation** • **procalcitonin** + clinical evaluation to help **stop** antibiotics (not to start) • **shorter courses** (e.g., 5–7 days) when source controlled.

Fluid resuscitation

- **Crystalloids first-line** • **balanced crystalloids (LR, Plasma-Lyte) preferred over 0.9% NaCl** (↓ hyperchloremic acidosis and AKI).
- **Albumin** in patients who received **large volumes of crystalloid**.
- **Avoid hydroxyethyl starches** (AKI, mortality, bleeding) and gelatins.
- **Reassess fluid responsiveness** with **dynamic measures** (passive leg raise, stroke volume/pulse pressure variation, fluid challenge) rather than static CVP alone • **capillary refill time** and lactate trends guide resuscitation.
- After resuscitation → **conservative fluids**; de-resuscitation (diuretics) if overloaded.

Vasopressors and inotropes in septic shock

- **Norepinephrine = first-line vasopressor** (0.01–3 mcg/kg/min; usual 2–100 mcg/min) • goal **MAP ≥ 65 mmHg**.
- **Add vasopressin 0.03 units/min** (fixed dose, not titrated) when **norepinephrine 0.25–0.5 mcg/kg/min** instead of escalating NE further (catecholamine-sparing).
- **Add epinephrine** if MAP still inadequate with NE + vasopressin.
- **Cardiac dysfunction with persistent hypoperfusion** despite adequate volume and MAP → **add dobutamine** or switch to epinephrine.
- **Dopamine**: only in highly selected patients (e.g., bradycardia, low arrhythmia risk) — **more arrhythmias** than NE; **low-dose "renal" dopamine not recommended**.
- **Angiotensin II** (20 ng/kg/min) for refractory vasodilatory shock (↑ thrombosis → VTE prophylaxis) • **phenylephrine** (pure α1) if tachyarrhythmias or as salvage.
- **Peripheral IV start is acceptable** (short term, proximal large vein) rather than delaying vasopressors until a central line • place **arterial line** when feasible.
- **Extravasation** of vasopressors → **phentolamine 5–10 mg in 10 mL NS** infiltrated SC into the area (α-blocker) • topical nitroglycerin, terbutaline alternatives.

Corticosteroids and other adjuncts

- **Hydrocortisone 200 mg/day** (**50 mg IV q6h** or continuous infusion) for septic shock with **ongoing vasopressor need** — **NE (or epi) ≥ 0.25 mcg/kg/min for ≥ 4 h** • ± **fludrocortisone 50 mcg PO daily** (APROCCHSS) • taper when vasopressors stopped • ADRs: hyperglycemia, hypernatremia, neuromuscular weakness.
- **Not recommended:** IV vitamin C, **sodium bicarbonate for lactic acidosis** (unless **pH ≤ 7.2 and AKI stage 2–3**), IVIG, polymyxin B hemoperfusion, erythropoietin, antithrombin.
- **Blood transfusion:** restrictive — **Hb < 7 g/dL** (unless MI, active bleeding, severe hypoxemia).
- **Glucose:** start insulin when glucose **≥ 180 mg/dL** → target **144–180**.
- **VTE prophylaxis:** pharmacologic — **LMWH preferred over UFH** (unless contraindicated) + mechanical if bleeding risk.
- **Stress ulcer prophylaxis** (PPI or H2RA) only with **risk factors for GI bleeding** (mechanical ventilation > 48 h, coagulopathy, shock, history of GI bleed, high-dose steroids + NSAID).
- **ARDS: low tidal volume 6 mL/kg predicted body weight**, **plateau pressure ≤ 30 cm H2O**, higher PEEP for moderate–severe, **prone positioning > 12 h/day** (P/F < 150), NMB boluses if needed, conservative fluids.
- **Nutrition: early enteral nutrition within 72 h • renal replacement** for usual indications (not just for creatinine rise) • goals-of-care discussion within 72 h.

4. Shock

Shock = circulatory failure → inadequate tissue oxygen delivery → cellular hypoxia (↑ **lactate > 2**) • signs: **hypotension** (SBP < 90 or MAP < 65), tachycardia, **altered mental status**, **oliguria** (< 0.5 mL/kg/h), cool mottled skin (warm in early distributive), delayed capillary refill.

Hemodynamic parameters

Parameter	Normal	Meaning
MAP = (SBP + 2 × DBP)/3	70–105 mmHg (goal **≥ 65**)	Organ perfusion pressure
Cardiac output (CO) = HR × SV	4–7 L/min	Pump function
Cardiac index (CI) = CO/BSA	2.5–4.2 L/min/m²	< 2.2 in cardiogenic shock
PCWP (wedge, left preload)	6–12 mmHg	↑ in cardiogenic, ↓ in hypovolemic
CVP (right preload)	2–6 mmHg (ventilated 8–12)	Volume status (poor predictor of fluid response)
SVR (afterload)	800–1,200 dyn·s/cm⁵	↓ in distributive, ↑ in hypovolemic/cardiogenic/obstructive
ScvO2 / SvO2	≥ 70% / 65–75%	↓ = ↑ oxygen extraction (low CO); ↑ in sepsis (poor extraction)
Lactate	< 2 mmol/L	Tissue hypoperfusion

Type	CO / CI	PCWP	SVR	Causes
Hypovolemic	↓	↓	↑	**Hemorrhage** (trauma, GI bleed), vomiting/diarrhea, burns, diuresis, third spacing (pancreatitis)
Cardiogenic	↓↓	↑	↑	**Acute MI** (most common), acute HF, arrhythmias, myocarditis, valvular disease
Distributive	**↑ or normal**	↓/normal	↓↓	**Sepsis** (most common shock overall), **anaphylaxis**, **neurogenic** (spinal cord injury), adrenal crisis, liver failure
Obstructive	↓	Variable (↑ CVP)	↑	**Massive PE**, **cardiac tamponade** (Beck's triad), **tension pneumothorax**, severe pulmonary HTN

Management by type

- **General:** airway/oxygen, IV access, **MAP ≥ 65**, treat cause, monitor lactate, urine output, mental status • **VIP**: ventilate, infuse, pump.
- **Hypovolemic (non-hemorrhagic): isotonic crystalloids** (LR/NS boluses 500–1,000 mL) • albumin 5% optional • avoid starches.
- **Hemorrhagic: stop the bleeding** (surgery/endoscopy/IR) • **blood products early** — **massive transfusion 1:1:1** (plasma : platelets : RBCs) • **limit crystalloids** (dilutional coagulopathy, acidosis, hypothermia — "lethal triad") • **tranexamic acid 1 g IV over 10 min then 1 g over 8 h within 3 h** of trauma • permissive hypotension (SBP 80–90) in penetrating trauma until hemorrhage control (not in TBI) • calcium replacement (citrate in blood products → hypocalcemia) • reverse anticoagulants.
- **Cardiogenic: revascularization (PCI)** for MI • **norepinephrine** for hypotension (preferred over dopamine — SOAP II: dopamine ↑ arrhythmias and mortality in cardiogenic) • **inotropes** for low CO: **dobutamine** (β1 → ↑ contractility; ↑ HR, arrhythmias, β2-vasodilation → hypotension) or **milrinone** (PDE3 inhibitor → inodilator; **hypotension**, arrhythmias, **renal dose adjustment**; works with β-blockers) • **diuretics** if congested (avoid fluids if PCWP high) • mechanical circulatory support (**IABP**, Impella, VA-ECMO) • avoid negative inotropes (β-blockers, non-DHP CCBs) acutely.
- **Septic:** see Section 3.
- **Anaphylactic: epinephrine IM 0.3–0.5 mg (0.01 mg/kg; 1 mg/mL)** in **anterolateral thigh**, repeat every 5–15 min → IV epinephrine infusion if refractory • **large-volume fluids** • adjuncts (not substitutes): H1 (diphenhydramine 25–50 mg), H2 (famotidine 20 mg), **albuterol** (bronchospasm), corticosteroids (methylprednisolone 1–2 mg/kg — prevent biphasic reaction, not proven) • **patients on β-blockers → glucagon 1–5 mg IV over 5 min**, then infusion 5–15 mcg/min.
- **Neurogenic** (spinal cord injury above T6 → loss of sympathetic tone): **hypotension + bradycardia**, warm skin → fluids, **norepinephrine** (α + β1), **atropine**/pacing for bradycardia • avoid phenylephrine alone (reflex bradycardia) • MAP goal 85–90 mmHg for ~7 days.
- **Adrenal crisis: hydrocortisone 100 mg IV bolus** then 50 mg q6h (200 mg/day) + dextrose-saline fluids.
- **Obstructive: PE → systemic thrombolysis** (alteplase 100 mg over 2 h) or embolectomy • **tamponade → pericardiocentesis** • **tension pneumothorax → needle decompression** then chest

tube • fluids as bridge.

Vasopressors and inotropes — receptor pharmacology

Drug	Receptors	Dose	Effects / ADRs
Norepinephrine	**α1 >> β1**	0.01–3 mcg/kg/min (2–100 mcg/min)	↑ SVR, ↑ MAP, modest ↑ CO • **first-line** septic, cardiogenic, most shock • extravasation/ischemia
Epinephrine	**β1, β2, α1** (α at higher doses)	0.01–2 mcg/kg/min (1–40 mcg/min)	↑ CO and SVR • **tachyarrhythmias**, ↑ **lactate** (β2 — aerobic glycolysis), hyperglycemia, splanchnic ischemia • anaphylaxis, 2nd/3rd-line septic shock
Vasopressin	**V1** (vascular)	**0.03–0.04 units/min fixed**	Non-adrenergic vasoconstriction (works in acidosis) • NE-sparing • digital/mesenteric ischemia, hyponatremia (V2)
Phenylephrine	**Pure α1**	0.5–9 mcg/kg/min (10–200 mcg/min)	↑ SVR, **reflex bradycardia**, may ↓ CO • used if tachyarrhythmias, anesthesia hypotension
Dopamine	Dose-dependent: **D1** (1–3), **β1** (3–10), **α1** (> 10 mcg/kg/min)	2–20 mcg/kg/min	**Arrhythmias** (tachycardia, AF) • bradycardia with hypotension • renal dose not effective
Dobutamine	**β1 > β2** (mild α1)	2–20 mcg/kg/min	**Inotrope**: ↑ CO, ↓ SVR → may ↓ BP • tachycardia, arrhythmias, ischemia • tachyphylaxis (> 72 h)
Milrinone	**PDE3 inhibitor** (↑ cAMP)	0.125–0.75 mcg/kg/min (no bolus usually)	**Inodilator**: ↑ CO, ↓ SVR, ↓ PVR • **hypotension**, arrhythmias • **renally cleared — ↓ dose in CKD** • long half-life • effective with β-blockers
Angiotensin II	**AT1**	20 ng/kg/min (max 80 first 3 h, then 40)	Refractory vasodilatory shock • **thrombosis** (give VTE prophylaxis), tachycardia
Isoproterenol	β1 = β2	2–10 mcg/min	Bradycardia, torsades (↑ HR) • vasodilation, arrhythmias

5. Cardiac Arrest — Adult ACLS (AHA)

High-quality CPR

- **Chain of survival:** early recognition and activation → **immediate CPR** → **rapid defibrillation** → advanced care → post-arrest care → recovery.
- **Compressions:** rate **100–120/min**, depth **at least 2 in (5 cm) but not > 2.4 in (6 cm)**, allow **full chest recoil**, **minimize interruptions** (chest compression fraction > 60–80%; pauses < 10 s), avoid excessive ventilation • switch compressor every 2 min.
- **Ventilation: 30:2** without advanced airway • **with advanced airway: 1 breath every 6 seconds (10/min)** with continuous compressions.
- **Monitor quality: ETCO2** — **< 10 mmHg** = poor CPR quality (and after 20 min, poor prognosis) • **abrupt ↑ to ≥ 35–40 mmHg suggests ROSC** • arterial diastolic pressure ≥ 20–25 mmHg.
- **Drug route: IV first** (peripheral); **IO** if IV unsuccessful/not feasible • follow each drug with a **20 mL flush** and raise the extremity • **endotracheal** (last resort, **2–2.5× dose**): **NAVEL** — naloxone, atropine, vasopressin, epinephrine, lidocaine.

Shockable rhythms — VF / pulseless VT

1. Start CPR, give oxygen, attach monitor/defibrillator → **shock** (biphasic **120–200 J** per manufacturer, else max; monophasic **360 J**).
2. **CPR 2 min** → IV/IO access → rhythm check → **shock**.
3. **CPR 2 min** → **epinephrine 1 mg IV/IO every 3–5 min** (after the 2nd shock) → consider advanced airway, capnography → rhythm check → **shock**.

4. **CPR 2 min → amiodarone 300 mg IV/IO bolus** (then **150 mg** for 2nd dose) **or lidocaine 1–1.5 mg/kg** (then 0.5–0.75 mg/kg; max 3 mg/kg) → treat reversible causes.
5. Refractory VF: consider **double sequential defibrillation** or vector change, ECPR in selected centers.

Non-shockable rhythms — PEA / asystole

- **CPR 2 min cycles** + **epinephrine 1 mg IV/IO as soon as possible**, then every 3–5 min • **no shocks** • **atropine is NOT recommended** for PEA/asystole.
- **Search for and treat reversible causes (H's and T's)** — PEA is usually caused by one.
- Check rhythm every 2 min → if organized rhythm → pulse check → if ROSC → post-cardiac arrest care.

H's	T's
Hypovolemia → fluids/blood	**Tension pneumothorax** → needle decompression
Hypoxia → oxygen, airway	**Tamponade** (cardiac) → pericardiocentesis
Hydrogen ion (acidosis) → ventilation; bicarbonate in selected cases	**Toxins** → specific antidotes
Hypo-/Hyperkalemia → hyperK: **calcium chloride 1 g** (or gluconate 3 g) + bicarbonate + insulin/dextrose	**Thrombosis, pulmonary** (PE) → **thrombolytics** (alteplase 50 mg IV bolus, may repeat; continue CPR 60–90 min)
Hypothermia → rewarming	**Thrombosis, coronary** (MI) → PCI after ROSC

Other arrest drugs and special situations

- **Not recommended routinely: vasopressin** (no advantage over epinephrine), **sodium bicarbonate**, **calcium**, magnesium (except torsades), atropine.
- **Torsades de pointes: magnesium sulfate 1–2 g IV** (bolus in arrest; over 5–20 min with pulse) • stop QT-prolonging drugs • overdrive pacing/isoproterenol.
- **Hyperkalemia arrest: calcium chloride 10% 10 mL (1 g)** IV/IO (stabilizes myocardium) + sodium bicarbonate 50 mEq + insulin 10 units + dextrose 25 g.
- **Opioid overdose:** CPR is the priority in arrest • **naloxone** (2 mg IN or 0.4 mg IM/IV) for respiratory arrest with pulse — may be given in arrest.
- **Local anesthetic systemic toxicity (LAST): 20% lipid emulsion 1.5 mL/kg bolus** then 0.25 mL/kg/min • reduce epinephrine doses (≤ 1 mcg/kg).
- **TCA overdose** (wide QRS) → **sodium bicarbonate** • **β-blocker / CCB overdose** → glucagon, calcium, high-dose insulin euglycemia therapy, lipid emulsion • **digoxin** → digoxin immune Fab • **cyanide** → hydroxocobalamin.
- **Pregnancy:** manual left uterine displacement • perimortem cesarean delivery within ~5 min if no ROSC (≥ 20 weeks) • standard drug doses.
- **Pediatric:** compressions 1/3 AP diameter (~2 in child, 1.5 in infant), 15:2 with 2 rescuers • **epinephrine 0.01 mg/kg** (0.1 mL/kg of 0.1 mg/mL; max 1 mg) • **defibrillation 2 J/kg → 4 J/kg** (max 10 J/kg or adult dose) • **amiodarone 5 mg/kg** (up to 3 doses) or lidocaine 1 mg/kg.

Bradycardia and tachycardia with a pulse (peri-arrest)

- **Symptomatic bradycardia** (hypotension, altered mental status, shock, chest pain, acute HF): **atropine 1 mg IV every 3–5 min (max 3 mg)** → if ineffective: **transcutaneous pacing** or **dopamine 5–20 mcg/kg/min** or **epinephrine 2–10 mcg/min** infusion → transvenous pacing • atropine ineffective in heart transplant, high-degree AV block (Mobitz II, 3rd degree) → pace.
- **Unstable tachycardia** (hypotension, altered mental status, shock, ischemic chest pain, acute HF) → **synchronized cardioversion** (sedate if possible) • polymorphic VT without pulse → **unsynchronized defibrillation**.
- **Stable narrow regular (SVT): vagal maneuvers** (modified Valsalva) → **adenosine 6 mg rapid IV push + 20 mL flush → 12 mg** if needed • ↓ dose (3 mg) via central line, heart transplant, or with **dipyridamole/carbamazepine** • ↑ dose needed with **theophylline/caffeine** (antagonists) • ADRs: transient asystole, flushing, chest pressure, bronchospasm (caution asthma) • then β-blocker or diltiazem.
- **Stable wide QRS (monomorphic VT): amiodarone 150 mg IV over 10 min** (repeat; then 1 mg/min × 6 h) or **procainamide** 20–50 mg/min (avoid with QT/HF) or sotalol • adenosine only if regular and monomorphic.

- **AF with RVR:** rate control (diltiazem 0.25 mg/kg IV, metoprolol 2.5–5 mg IV) • amiodarone/digoxin in HF/hypotension • cardioversion if unstable.

Post-cardiac arrest care (after ROSC)

- **Airway/oxygen:** SpO2 **92–98%** (avoid hypoxia and hyperoxia) • PaCO2 **35–45 mmHg**.
- **Hemodynamics:** avoid hypotension — **SBP ≥ 90**, **MAP ≥ 65 mmHg** (fluids, norepinephrine/epinephrine).
- **12-lead ECG → emergent PCI for STEMI** (or suspected cardiac cause with shock/electrical instability).
- **Temperature control** for comatose patients (not following commands): maintain a target **32–37.5 °C** and **actively prevent fever** for at least **72 h** (prophylactic antipyretics, cooling devices) • manage shivering (meperidine, buspirone, sedation, NMB).
- **Glucose** control (avoid hypoglycemia; 140–180) • treat **seizures** (EEG) — no seizure prophylaxis • **neuroprognostication ≥ 72 h after return to normothermia** (multimodal) • avoid premature withdrawal of care.

6. Acid–Base Disorders

Normal values

Parameter	Normal
pH	**7.35–7.45** (< 7.35 acidemia, > 7.45 alkalemia)
PaCO2 (respiratory)	**35–45 mmHg** (40)
HCO3 (metabolic)	**22–26 mEq/L** (24)
PaO2	80–100 mmHg
Anion gap = Na − (Cl + HCO3)	**8–12 mEq/L** (~4–12 with newer analyzers)
Albumin-corrected AG	AG + **2.5 × (4 − albumin g/dL)** (each 1 g/dL ↓ albumin lowers AG by 2.5)
Serum osmolality (calculated)	2 × Na + glucose/18 + BUN/2.8 (275–295) • **osmolal gap** = measured − calculated (**< 10**)

Stepwise approach

1. **pH**: acidemia or alkalemia?
2. **Primary disorder**: does PaCO2 or HCO3 explain the pH? (pH ↓ + HCO3 ↓ = metabolic acidosis • pH ↓ + PaCO2 ↑ = respiratory acidosis • pH ↑ + HCO3 ↑ = metabolic alkalosis • pH ↑ + PaCO2 ↓ = respiratory alkalosis).
3. **Compensation**: is it appropriate? (if not → a **second (mixed) disorder**). **Compensation never overcorrects** the pH back to normal.
4. **Anion gap** (always calculate, even if pH normal) — correct for albumin.
5. If AG ↑ → **delta-delta**: ΔAG/ΔHCO3 = (AG − 12)/(24 − HCO3) • **< 1** = coexisting **non-AG metabolic acidosis** • **1–2** = pure AG acidosis • **> 2** = coexisting **metabolic alkalosis** (or chronic respiratory acidosis) • equivalently, corrected HCO3 = HCO3 + ΔAG (> 26 alkalosis, < 22 NAGMA).
6. If AG ↑ and toxic ingestion suspected → **osmolal gap** (> 10 → toxic alcohols).
7. Non-AG acidosis → **urine anion gap** to separate GI vs renal causes.

Expected compensation

Primary disorder	Expected compensation
Metabolic acidosis	**Winter's formula: expected PaCO2 = 1.5 × HCO3 + 8 (± 2)** • (quick check: PaCO2 ≈ last 2 digits of pH) • PaCO2 higher than expected → concurrent respiratory acidosis; lower → respiratory alkalosis
Metabolic alkalosis	PaCO2 ↑ **0.7 mmHg per 1 mEq/L ↑ HCO3** (expected PaCO2 ≈ 0.7 × HCO3 + 21 ± 2; rarely > 55)
Acute respiratory acidosis	HCO3 ↑ **1** per 10 mmHg ↑ PaCO2 (pH ↓ ~0.08 per 10)
Chronic respiratory acidosis (> 3–5 days)	HCO3 ↑ **3.5–4** per 10 mmHg ↑ PaCO2 (pH ↓ ~0.03 per 10)
Acute respiratory alkalosis	HCO3 ↓ **2** per 10 mmHg ↓ PaCO2
Chronic respiratory alkalosis	HCO3 ↓ **4–5** per 10 mmHg ↓ PaCO2

Metabolic acidosis — anion gap (AGMA)

- **GOLDMARK**: **G**lycols (**ethylene glycol**, **propylene glycol** — lorazepam, diazepam, phenobarbital IV vehicles), **O**xoproline (pyroglutamic acid — chronic **acetaminophen** in malnourished women), **L**-lactate, **D**-lactate (short bowel), **M**ethanol, **A**spirin (salicylates), **R**enal failure (uremia), **K**etoacidosis (diabetic, alcoholic, starvation) • older mnemonic: **MUDPILES** (methanol, uremia, DKA, propylene glycol/paraldehyde, iron/INH, lactate, ethylene glycol, salicylates).
- **Lactic acidosis type A** (hypoperfusion/hypoxia): shock, sepsis, cardiac arrest, severe anemia, CO poisoning, mesenteric ischemia.
- **Lactic acidosis type B** (no hypoperfusion): **metformin**, **linezolid**, **propofol (PRIS)**, **NRTIs** (zidovudine, stavudine, didanosine), **epinephrine/albuterol** (β2), **nitroprusside (cyanide)**, isoniazid, **alcohol**, liver failure, malignancy, thiamine deficiency, seizures.
- **Toxic alcohols** (methanol → blindness; ethylene glycol → **AKI, calcium oxalate crystals**, hypocalcemia): ↑ **osmolal gap** early → ↑ AG later • isopropanol → ↑ osmolal gap + **ketosis without acidosis**.
- **Salicylate toxicity: mixed respiratory alkalosis (early) + AG metabolic acidosis**, tinnitus, hyperthermia, N/V.

Metabolic acidosis — normal anion gap (hyperchloremic, NAGMA)

- **GI loss of HCO3: diarrhea** (most common), pancreatic/biliary fistulas, ureteral diversion (ileal conduit), laxative abuse.
- **Renal tubular acidosis (RTA): • Type 1 (distal)** — can't excrete H+ → **hypokalemia**, urine pH **> 5.5**, kidney stones/nephrocalcinosis — **amphotericin B**, **lithium**, ifosfamide, autoimmune (Sjögren's) • **Type 2 (proximal)** — HCO3 wasting → **hypokalemia**, Fanconi syndrome — **acetazolamide**, **topiramate**, **tenofovir**, ifosfamide, aminoglycosides, multiple myeloma • **Type 4** (hypoaldosteronism) — **hyperkalemia** — diabetes, **ACEi/ARB, spironolactone/eplerenone, amiloride, trimethoprim, heparin, NSAIDs, calcineurin inhibitors**.
- **Iatrogenic: large-volume 0.9% NaCl** (dilutional/hyperchloremic), **carbonic anhydrase inhibitors**, cholestyramine, **excess chloride in PN** (use acetate), early CKD.
- **Urine anion gap** = U-Na + U-K − U-Cl • **negative** ("neGUTive") = appropriate NH4+ excretion → **GI loss** (diarrhea) • **positive** → **RTA** (types 1, 4).

Treatment of metabolic acidosis

- **Treat the underlying cause** (fluids/insulin in DKA, perfusion in lactic acidosis, dialysis in uremia, stop offending drug).
- **IV sodium bicarbonate** is **controversial in AG acidosis** (lactic, ketoacidosis) — consider when **pH < 7.1–7.2** (esp. with **AKI**), DKA with pH < 7.0, **hyperkalemia**, or **specific toxins** (salicylates, TCAs, methanol) • more appropriate for **NAGMA** (true HCO3 loss).
- **Bicarbonate deficit (mEq) = 0.5 × weight (kg) × (desired HCO3 − measured HCO3)** → give **~half over 3–4 h**, reassess • 8.4% = 1 mEq/mL; isotonic 150 mEq/L in D5W as infusion.
- **Bicarbonate ADRs: hypernatremia**, **volume overload**, **hypokalemia** (K shifts into cells), **↓ ionized calcium**, **↑ CO2 production** (paradoxical intracellular/CSF acidosis — needs adequate ventilation), overshoot alkalosis, tissue injury if extravasated.

- **THAM (tromethamine)**: non-sodium buffer that doesn't generate CO2 (hypernatremia, hypercapnia) • avoid in renal failure, hyperkalemia.
- **Oral alkali (chronic — CKD, RTA): sodium bicarbonate tablets** (650 mg = 7.7 mEq; bloating/gas), **sodium citrate** (Bicitra, 1 mEq/mL), **potassium citrate** (RTA with hypokalemia/stones) • CKD goal HCO3 ≥ 22.
- **Toxic alcohols: fomepizole** (alcohol dehydrogenase inhibitor) **15 mg/kg IV load, then 10 mg/kg q12h × 4 doses, then 15 mg/kg q12h** (autoinduction) • ↑ frequency to q4h during dialysis • ethanol as alternative • **hemodialysis** (severe acidosis, AKI, visual changes, high levels) • cofactors: **folic acid/leucovorin** (methanol), **thiamine + pyridoxine** (ethylene glycol).
- **Salicylates: urine alkalinization** with sodium bicarbonate (goal urine pH 7.5–8; keep K normal), dialysis if severe • avoid intubation if possible (loss of compensatory hyperventilation).
- **Metformin-associated lactic acidosis:** supportive + **hemodialysis** • **propylene glycol:** stop lorazepam/IV diazepam, dialysis if severe.

Metabolic alkalosis

Type	Urine chloride	Causes	Treatment
Saline (chloride)-responsive	**< 20 mEq/L** (often < 10)	**Vomiting**, **NG suction**, **remote diuretic use**, contraction alkalosis, post-hypercapnia, villous adenoma, cystic fibrosis	**0.9% NaCl + KCl** • correct Mg • **PPI/H2RA** to ↓ acid loss from NG suction • antiemetics
Saline-resistant	**> 20 mEq/L**	**Mineralocorticoid excess** (primary hyperaldosteronism, Cushing's, licorice — with **HTN**), **current diuretic use**, severe **hypokalemia** (< 2), **hypomagnesemia**, Bartter/Gitelman	**Spironolactone**/eplerenone or **amiloride**, treat cause (surgery), **replace K and Mg**
Alkali load	Variable	**Massive transfusion (citrate)**, **acetate in PN**, bicarbonate/antacids, **milk-alkali syndrome** (Ca carbonate), lactate in large LR volumes (with renal impairment)	Stop source • switch PN acetate → chloride

- Maintained by **volume depletion**, **hypokalemia**, **chloride depletion**, and ↑ aldosterone (kidney keeps reabsorbing HCO3).
- Symptoms: muscle cramps, weakness, **arrhythmias**, paresthesias/tetany (↓ ionized Ca), hypoventilation, confusion; ↑ digoxin toxicity with hypokalemia.
- **Acetazolamide 250–500 mg IV/PO** (↑ renal HCO3 excretion): for alkalosis with **volume overload** (HF with diuretics, post-hypercapnic) • causes hypokalemia, NAGMA, sulfonamide allergy caution.
- **Hydrochloric acid 0.1 N** via **central line** (or ammonium chloride — avoid in liver/renal failure) for **severe** alkalosis (pH > 7.55–7.6) with life-threatening symptoms • dialysis (low-HCO3 bath) in renal failure.

Respiratory acidosis (hypoventilation, ↑ PaCO2)

- **Causes: COPD/asthma exacerbation**, obesity hypoventilation, sleep apnea, **CNS depression** (**opioids**, **benzodiazepines**, barbiturates, propofol, alcohol, stroke), **neuromuscular** (**myasthenia gravis**, **Guillain-Barré**, ALS, NMB, hypokalemia/hypophosphatemia), chest wall (kyphoscoliosis, flail chest), pneumothorax, **excess O2 in chronic CO2 retainers**, ventilator settings, high-carbohydrate feeds (↑ CO2).
- Symptoms: headache, somnolence, confusion, **CO2 narcosis**, asterixis, vasodilation, ↑ ICP.
- **Treatment:** treat cause • **ventilatory support** — **NIPPV/BiPAP** (COPD) or intubation • **naloxone** (opioids), flumazenil (benzodiazepines — seizure risk, rarely) • bronchodilators/steroids • target **SpO2 88–92%** in COPD • **avoid rapid correction of chronic hypercapnia** (post-hypercapnic metabolic alkalosis, seizures) • bicarbonate not indicated.

Respiratory alkalosis (hyperventilation, ↓ PaCO2)

- **Causes: anxiety/pain**, **hypoxemia** (**PE**, pneumonia, asthma, altitude, HF), **early sepsis**, **salicylates** (early), **pregnancy** and **progesterone**, **liver failure**, CNS disease (stroke, tumor, infection), fever, hyperthyroidism, **mechanical overventilation**, theophylline/methylxanthines, catecholamines.

- Symptoms: lightheadedness, **perioral numbness**, **paresthesias**, carpopedal spasm/**tetany** (↓ ionized Ca), arrhythmias, ↓ cerebral blood flow.
- **Treatment:** treat underlying cause (oxygen, analgesia, anxiolysis) • ↓ ventilator rate/tidal volume • **paper-bag rebreathing not recommended** (risk of hypoxemia).

Drug-induced acid–base disorders (summary)

Disorder	Drugs
AG metabolic acidosis	Metformin, linezolid, propofol, NRTIs, salicylates, **propylene glycol** (lorazepam, diazepam, phenobarbital, etomidate IV), acetaminophen (5-oxoproline), methanol/ethylene glycol, SGLT2i (ketoacidosis), epinephrine, nitroprusside (cyanide), isoniazid, iron
NAG metabolic acidosis	**Acetazolamide, topiramate, zonisamide**, **amphotericin B**, **tenofovir**, ifosfamide, lithium, **spironolactone**, amiloride, **trimethoprim**, ACEi/ARB, NSAIDs, heparin, calcineurin inhibitors, cholestyramine, **large-volume NS**, PN chloride
Metabolic alkalosis	**Loop and thiazide diuretics**, **corticosteroids/fludrocortisone**, **bicarbonate/citrate/acetate/lactate** loads, **massive transfusion**, **antacids + Ca carbonate (milk-alkali)**, penicillins in high doses (nonabsorbable anions), licorice
Respiratory acidosis	**Opioids, benzodiazepines**, barbiturates, propofol, **neuromuscular blockers**, aminoglycosides (NM blockade), excess O2 in COPD
Respiratory alkalosis	**Salicylates (early)**, progesterone, methylxanthines (theophylline), catecholamines, nicotine

Worked example: pH 7.25, PaCO2 28, HCO3 12, Na 140, Cl 104, albumin 4 → **acidemia** + ↓ HCO3 = **metabolic acidosis** • AG = 140 − (104 + 12) = **24 (↑)** • Winter's: 1.5 × 12 + 8 = 26 ± 2 → measured 28 = **appropriate** • ΔAG/ΔHCO3 = (24 − 12)/(24 − 12) = **1** → **pure AG metabolic acidosis** (e.g., DKA or lactic acidosis).

7. Acute Medical Diseases — Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Fentanyl	0.35–0.5 mcg/kg IV q30–60 min • infusion 0.7–10 mcg/kg/h	Respiratory depression, **chest wall rigidity**, constipation	Hemodynamically stable • renal OK • **CYP3A4** (azoles, ritonavir ↑) • serotonin syndrome
Hydromorphone	0.2–0.6 mg IV q1–2h • 0.5–3 mg/h	Respiratory depression, sedation	Potent (dose errors) • preferred over morphine in renal impairment
Morphine	2–4 mg IV q1–2h • 2–30 mg/h	**Histamine → hypotension**, pruritus	**Avoid in renal failure** (M6G/M3G)
Remifentanil	0.5–15 mcg/kg/h	Bradycardia, hyperalgesia after stop	**Esterase** metabolism — no accumulation
Acetaminophen IV	1 g q6h or 650 mg q4h (max 4 g; 2 g liver disease)	Hepatotoxicity, hypotension (IV)	Opioid-sparing • warfarin INR ↑ (regular use)
Ketamine	Analgesia 0.5 mg/kg → 1–2 mcg/kg/min • sedation 0.05–0.4 mg/kg/h	**Emergence reactions**, ↑ HR/BP, secretions	Preserves BP and respiratory drive • bronchodilator
Naloxone	0.04–0.4 mg IV titrated (2 mg IN/IM in overdose)	Withdrawal, pain, pulmonary edema, HTN	Short half-life → repeat/infusion
Propofol	**5–50 mcg/kg/min** (no bolus in ICU)	**Hypotension**, ↑ TG, pancreatitis, **PRIS**	1.1 kcal/mL • check TG at 48 h • no analgesia • change tubing 12 h
Dexmedetomidine	**0.2–1.5 mcg/kg/h**	**Bradycardia, hypotension**	No respiratory depression • arousable • ↓ delirium • not for seizures/deep sedation
Midazolam	0.01–0.05 mg/kg • 0.02–0.1 mg/kg/h	Respiratory depression, delirium	**CYP3A4** • active metabolite accumulates in renal failure

Drug	Dose	Key ADRs	Key notes / interactions
Lorazepam	0.02–0.04 mg/kg q2–6h • 0.01–0.1 mg/kg/h	Delirium, **propylene glycol toxicity**	Osmolal gap > 10–12 → stop • alcohol withdrawal, seizures
Phenobarbital	Alcohol withdrawal 10 mg/kg IV (or 130–260 mg q15–30 min)	Sedation, respiratory depression, hypotension	Strong **inducer** • propylene glycol
Haloperidol	0.5–5 mg IV q20–30 min PRN	**QTc/torsades**, EPS, NMS	Only for distressing delirium symptoms • hold if QTc > 500
Quetiapine	25–50 mg BID (max 200 mg BID)	Sedation, orthostasis, QT	Stop when delirium resolves
Cisatracurium	0.1–0.2 mg/kg → 1–3 mcg/kg/min	ICU-acquired weakness	**Hofmann elimination** • sedate first • TOF • eye care
Rocuronium	0.6–1.2 mg/kg (RSI)	Anaphylaxis (rare)	**Sugammadex** reversal (↓ hormonal contraceptives 7 days)
Succinylcholine	1–1.5 mg/kg IV	**Hyperkalemia**, **malignant hyperthermia**, bradycardia	CI burns, crush, immobility, NM disease • MH → dantrolene
Regular insulin IV (DKA)	**0.1 units/kg/h** (± 0.1 units/kg bolus) → 0.05 when BG < 250	**Hypokalemia**, hypoglycemia	**Hold if K < 3.5** • SC basal 1–2 h before stopping
Potassium chloride IV	10–20 mEq/h (peripheral ≤ 10)	Hyperkalemia, phlebitis	Never IV push • DKA if K < 5.0
Sodium bicarbonate	DKA pH < 7.0: 100 mEq over 2 h • arrest/TCA 1–2 mEq/kg • deficit 0.5 × wt × Δ	Hypernatremia, **hypokalemia**, ↓ iCa, ↑ CO2	Not routine in arrest/lactic acidosis • incompatible with Ca, catecholamines
Norepinephrine	**0.01–3 mcg/kg/min**	Arrhythmia, ischemia, extravasation	**1st-line shock** • MAP ≥ 65 • phentolamine for extravasation
Vasopressin	**0.03 units/min** fixed	Digital/mesenteric ischemia, hyponatremia	Add at NE 0.25–0.5 mcg/kg/min
Epinephrine	Shock 0.01–2 mcg/kg/min • **arrest 1 mg q3–5 min** • **anaphylaxis 0.3–0.5 mg IM**	Tachyarrhythmias, ↑ **lactate**, hyperglycemia	Bradycardia 2–10 mcg/min
Phenylephrine	0.5–9 mcg/kg/min	**Reflex bradycardia**, ↓ CO	Pure α1 • tachyarrhythmia patients
Dopamine	2–20 mcg/kg/min	**Arrhythmias**	Selected bradycardia • no renal dose
Dobutamine	2–20 mcg/kg/min	Tachycardia, arrhythmias, hypotension	Inotrope (cardiogenic, septic with low CO)
Milrinone	0.125–0.75 mcg/kg/min	**Hypotension**, arrhythmias	**Renal dose adjustment** • works with β-blockers
Angiotensin II	20 ng/kg/min	**Thrombosis**, tachycardia	Refractory vasodilatory shock • VTE prophylaxis
Hydrocortisone	**50 mg IV q6h** (200 mg/day) • adrenal crisis 100 mg bolus	Hyperglycemia, hypernatremia, weakness	NE ≥ 0.25 mcg/kg/min ≥ 4 h • ± fludrocortisone 50 mcg
Vancomycin	Load 20–35 mg/kg → AUC 400–600	**Nephrotoxicity**, infusion reaction	MRSA risk • ↑ AKI with pip-tazo
Pip-tazo	4.5 g q6h (extended infusion 3.375–4.5 g q8h over 4 h)	Hypersensitivity, ↓ platelets, AKI (with vanco)	Anti-pseudomonal
Cefepime	2 g q8h	**Neurotoxicity** (renal failure)	Renal adjust
Meropenem	1–2 g q8h (extended infusion)	Seizures (less than imipenem)	**↓ valproate levels** • ESBL

Drug	Dose	Key ADRs	Key notes / interactions
Amiodarone	**Arrest 300 mg → 150 mg** • VT with pulse 150 mg over 10 min → 1 mg/min	Hypotension, bradycardia, QT, phlebitis	Inhibits CYP2C9/3A4/P-gp (warfarin, digoxin ↑)
Lidocaine	1–1.5 mg/kg → 0.5–0.75 mg/kg (max 3 mg/kg)	CNS toxicity, seizures	Alternative to amiodarone
Atropine	**1 mg IV q3–5 min (max 3 mg)**	Tachycardia, anticholinergic	Symptomatic bradycardia • **not for PEA/asystole**
Adenosine	**6 mg rapid push → 12 mg**	Transient asystole, flushing, chest pressure, bronchospasm	↓ dose with dipyridamole, carbamazepine, central line • theophylline/caffeine ↓ effect
Magnesium sulfate	1–2 g IV (torsades)	Hypotension, flushing	Torsades only
Calcium chloride 10%	1 g (10 mL) IV slow/push in arrest	**Vesicant**, bradycardia	Hyperkalemia, CCB overdose • central line preferred
Alteplase	PE shock: 100 mg over 2 h • arrest PE: 50 mg bolus	**Bleeding**, ICH, angioedema	Check contraindications
Tranexamic acid	1 g over 10 min → 1 g over 8 h (≤ 3 h of trauma)	Thrombosis, seizures (high dose)	Hemorrhagic shock
Glucagon	β-blocker toxicity/anaphylaxis 1–5 mg IV → 5–15 mcg/min	N/V, hyperglycemia	Refractory anaphylaxis on β-blockers
Phentolamine	5–10 mg in 10 mL NS SC	Hypotension	Vasopressor extravasation
Fomepizole	15 mg/kg → 10 mg/kg q12h × 4 → 15 mg/kg	Headache, nausea	Methanol/ethylene glycol • q4h during dialysis
Acetazolamide	250–500 mg IV/PO daily–BID	Hypokalemia, NAGMA, paresthesia	Metabolic alkalosis with volume overload • sulfonamide

Exam Traps and Frequently Tested Points

1. **Treat pain first** (analgesia-first) • **NRS** if able; **BPS > 5** or **CPOT > 2** if not • vital signs alone are not valid.
2. **Fentanyl**: no histamine release, safe in renal failure, chest wall rigidity • **morphine**: histamine, avoid in renal failure • **remifentanil**: esterase • **avoid meperidine**.
3. **RASS goal −2 to 0** (light sedation) • daily SAT + SBT • **propofol or dexmedetomidine over benzodiazepines**.
4. **Benzodiazepines preferred only for** alcohol/BZD withdrawal, seizures, deep sedation/amnesia with paralysis.
5. **Propofol:** hypotension, ↑ TG (check 48 h), pancreatitis, **1.1 kcal/mL**, **PRIS** (> 70 mcg/kg/min > 48 h → acidosis, rhabdo, hyperK, arrhythmia).
6. **Dexmedetomidine:** α2 agonist, no respiratory depression, arousable, bradycardia/hypotension, ↓ delirium, not for seizures.
7. **Lorazepam infusion → propylene glycol toxicity** (↑ osmolal gap, AG acidosis, AKI) • **midazolam**: CYP3A4, accumulates in renal failure.
8. **NMBs have no sedation or analgesia** • cisatracurium = Hofmann (organ independent) • succinylcholine → hyperkalemia, MH (dantrolene).
9. **CAM-ICU**: acute change/fluctuation + inattention + (altered LOC or disorganized thinking) • ICDSC ≥ 4 • **hypoactive delirium most common**.
10. **No drug for delirium prevention** • antipsychotics only short-term for distressing symptoms • **benzodiazepines worsen delirium** (except withdrawal) • haloperidol → QTc (hold > 500).
11. **ABCDEF bundle** • early mobility • sleep hygiene • reorientation, glasses, hearing aids.
12. **DKA:** glucose ≥ 200, **pH < 7.3 / HCO3 < 18**, **β-OHB ≥ 3** • **HHS:** glucose ≥ 600, effective osm > 300, minimal ketones.
13. **DKA order: fluids → check K** (< 3.5 → **hold insulin**) → **insulin 0.1 units/kg/h → dextrose at BG < 250** (HHS < 300) → bicarbonate only pH < 7.0 • phosphate only < 1.

14. DKA resolved: **β-OHB < 0.6 + pH ≥ 7.3 or HCO3 ≥ 18 • SC basal 1–2 h before stopping IV insulin** • K looks normal but total body K is depleted.
15. **Euglycemic DKA with SGLT2i • cerebral edema in children** • corrected Na = Na + 1.6 per 100 mg/dL glucose above 100.
16. **Sepsis = organ dysfunction (SOFA ≥ 2) • septic shock = vasopressor for MAP ≥ 65 + lactate > 2** despite fluids • **qSOFA not alone** for screening.
17. **Hour-1 bundle:** lactate, **blood cultures before antibiotics**, broad-spectrum antibiotics, **30 mL/kg crystalloid** (hypotension or lactate ≥ 4), vasopressors MAP ≥ 65.
18. **Antibiotics within 1 h** in shock/probable sepsis • MRSA coverage if risk • double gram-negative only if MDR risk • extended-infusion β-lactams • procalcitonin to stop.
19. **Balanced crystalloids > NS** • albumin after large volumes • **no starches** • dynamic measures (passive leg raise) to assess fluid responsiveness.
20. **Norepinephrine first** → **vasopressin 0.03 units/min** (NE 0.25–0.5) → epinephrine • dobutamine for cardiac dysfunction • dopamine only selected • peripheral start OK • phentolamine for extravasation.
21. **Hydrocortisone 200 mg/day** if NE ≥ 0.25 mcg/kg/min for ≥ 4 h • **no vitamin C**, no bicarbonate unless pH ≤ 7.2 + AKI.
22. **Sepsis supportive:** Hb < 7 transfusion • glucose 144–180 (start ≥ 180) • LMWH prophylaxis • SUP only if risk • ARDS 6 mL/kg PBW, plateau ≤ 30, prone.
23. **Shock profiles:** hypovolemic ↓CO ↓PCWP ↑SVR • **cardiogenic ↓CO ↑PCWP ↑SVR • distributive ↑CO ↓SVR** • obstructive ↓CO ↑SVR.
24. **Hemorrhagic shock:** 1:1:1 blood products, limit crystalloids, **TXA within 3 h**, calcium • **cardiogenic:** PCI, NE (not dopamine), dobutamine/milrinone, IABP.
25. **Anaphylaxis: epinephrine IM 0.3–0.5 mg thigh first** • antihistamines/steroids are adjuncts • β-blocker patients → **glucagon**.
26. **Neurogenic shock:** hypotension + **bradycardia** • NE, atropine • MAP 85–90.
27. **Milrinone:** PDE3, inodilator, hypotension, **renal adjustment** • **phenylephrine:** pure α1, reflex bradycardia • **epinephrine ↑ lactate**.
28. **CPR: 100–120/min**, **2–2.4 in**, full recoil, minimal pauses, **30:2** or **1 breath q6s** with advanced airway • **ETCO2 < 10** = poor CPR.
29. **VF/pVT:** shock → CPR → shock → **epinephrine 1 mg q3–5 min** → shock → **amiodarone 300 → 150 mg** or lidocaine 1–1.5 mg/kg.
30. **PEA/asystole: epinephrine ASAP**, no shock, **no atropine** • look for **H's and T's**.
31. **Torsades → magnesium 1–2 g • hyperkalemia → calcium chloride 1 g** + bicarbonate + insulin/dextrose • PE arrest → alteplase • LAST → lipid emulsion.
32. **Vasopressin, bicarbonate, calcium not routine** in arrest • IV first, IO second • ET route NAVEL (2–2.5× dose).
33. **Bradycardia: atropine 1 mg q3–5 min (max 3 mg)** → pacing / dopamine / epinephrine • **SVT:** vagal → **adenosine 6 → 12 mg** rapid push (↓ with dipyridamole/carbamazepine/central line; theophylline ↓ effect) • unstable → **synchronized cardioversion**.
34. **Post-ROSC:** SpO2 92–98%, PaCO2 35–45, **MAP ≥ 65 / SBP ≥ 90**, PCI for STEMI, **temperature 32–37.5 °C and prevent fever ≥ 72 h**, prognosticate ≥ 72 h.
35. **Acid–base:** pH 7.35–7.45, PaCO2 35–45, HCO3 22–26, **AG = Na − (Cl + HCO3)** 8–12 • **correct AG for albumin (+2.5 per 1 g/dL)**.
36. **Winter's formula: PaCO2 = 1.5 × HCO3 + 8 ± 2** • metabolic alkalosis: ↑ 0.7 per 1 HCO3 • respiratory acute 1 (acidosis) / 2 (alkalosis) • chronic 3.5 / 4–5 per 10 mmHg.
37. **Delta-delta** < 1 → + NAGMA • > 2 → + metabolic alkalosis • **osmolal gap > 10** → toxic alcohols (fomepizole).
38. **AGMA = GOLDMARK • type B lactic acidosis drugs:** metformin, linezolid, propofol, NRTIs, epinephrine, nitroprusside.
39. **NAGMA:** diarrhea, RTA, NS • **urine AG negative = GI**, positive = RTA • RTA 1/2 → hypoK (amphotericin, acetazolamide, topiramate, tenofovir) • **RTA 4 → hyperK** (ACEi/ARB, spironolactone, trimethoprim, heparin).
40. **Bicarbonate deficit = 0.5 × wt × (desired − measured)**, give half • ADRs: hypernatremia, hypokalemia, ↓ iCa, ↑ CO2 • not routine in lactic acidosis/DKA.
41. **Metabolic alkalosis:** urine Cl **< 20 = saline-responsive** (vomiting, NG, diuretics remote → NS + KCl, PPI) • **> 20 = saline-resistant** (hyperaldosteronism → spironolactone) • **acetazolamide** if volume

overloaded • HCl 0.1 N (central) if severe.

42. **Respiratory acidosis:** opioids, BZDs, COPD, neuromuscular → ventilate (BiPAP), naloxone, O2 88–92% in COPD • **respiratory alkalosis:** anxiety, PE, sepsis, salicylates (early), pregnancy, liver failure.
43. **Salicylate toxicity = respiratory alkalosis + AG metabolic acidosis** → urine alkalinization (bicarbonate), dialysis.

> **Note:** doses and cut-offs follow current SCCM PADIS, Surviving Sepsis Campaign 2021, AHA ACLS, and the 2024 ADA/EASD hyperglycemic crises consensus. If your course slides or reference use slightly different numbers, **use your course numbers for the exam**.